

A POWER BI DASHBOARD SHOWCASE

Loan Performance & Risk Analysis

Six interactive dashboards covering executive KPIs, portfolio composition, borrower income, delinquency & credit risk, portfolio risk, and machine learning.

EA

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Data Analyst | July 2026

About This Report

DATASET SNAPSHOT

10,000

Loans Analyzed

\$164M

Total Portfolio Exposure

\$16.36K

Average Loan Size

12.43%

Average Interest Rate

0.07%

Observed Default Rate

17.73

Average Debt-to-Income

Prepared using Excel, SQL, Python, and Power BI as part of an end-to-end lending portfolio analytics project. This document is a companion visual reference for the project's GitHub repository.

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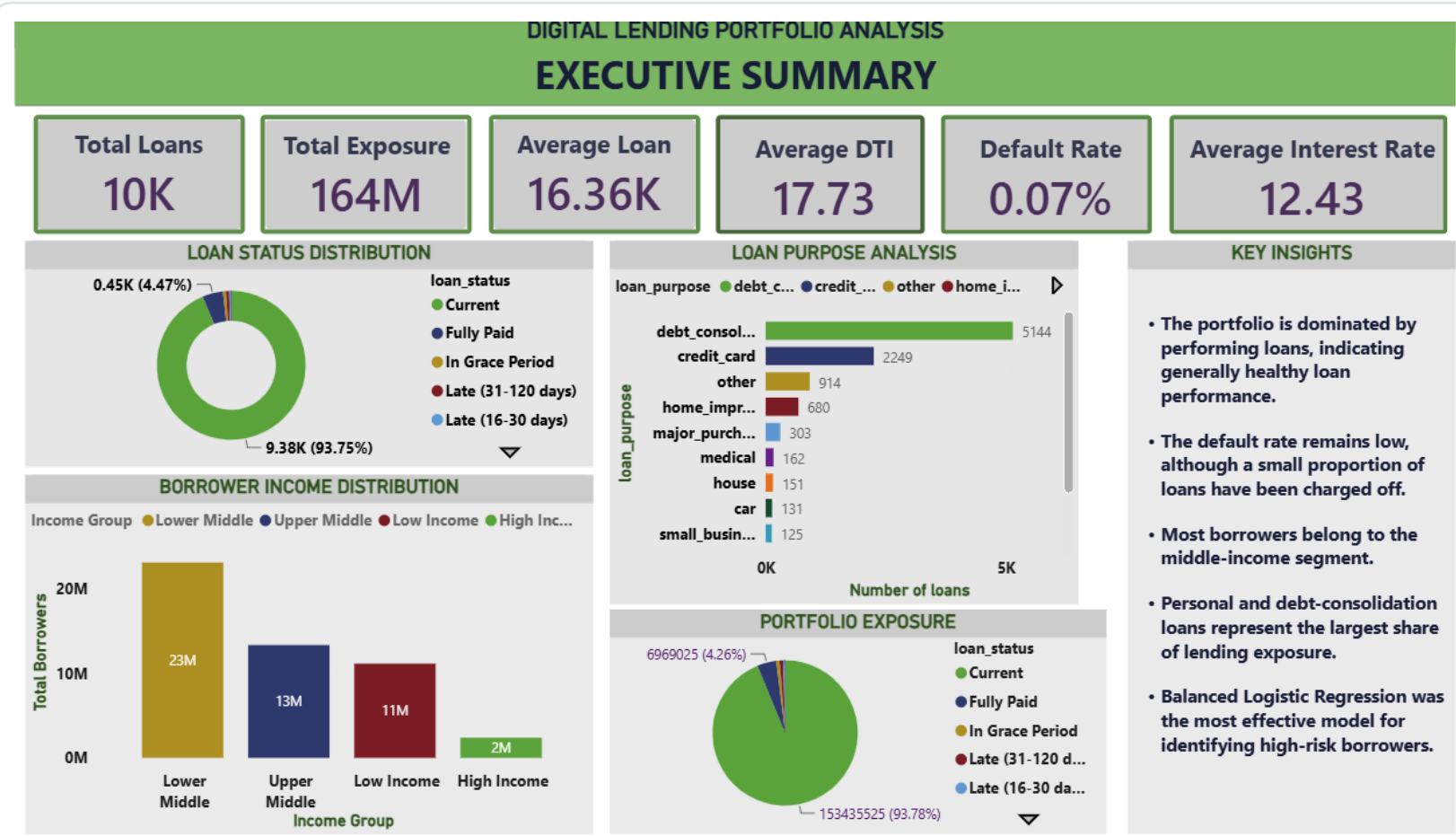
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Machine Learning Insights

Executive Summary

**93.75%**

Current Loans

0.07%

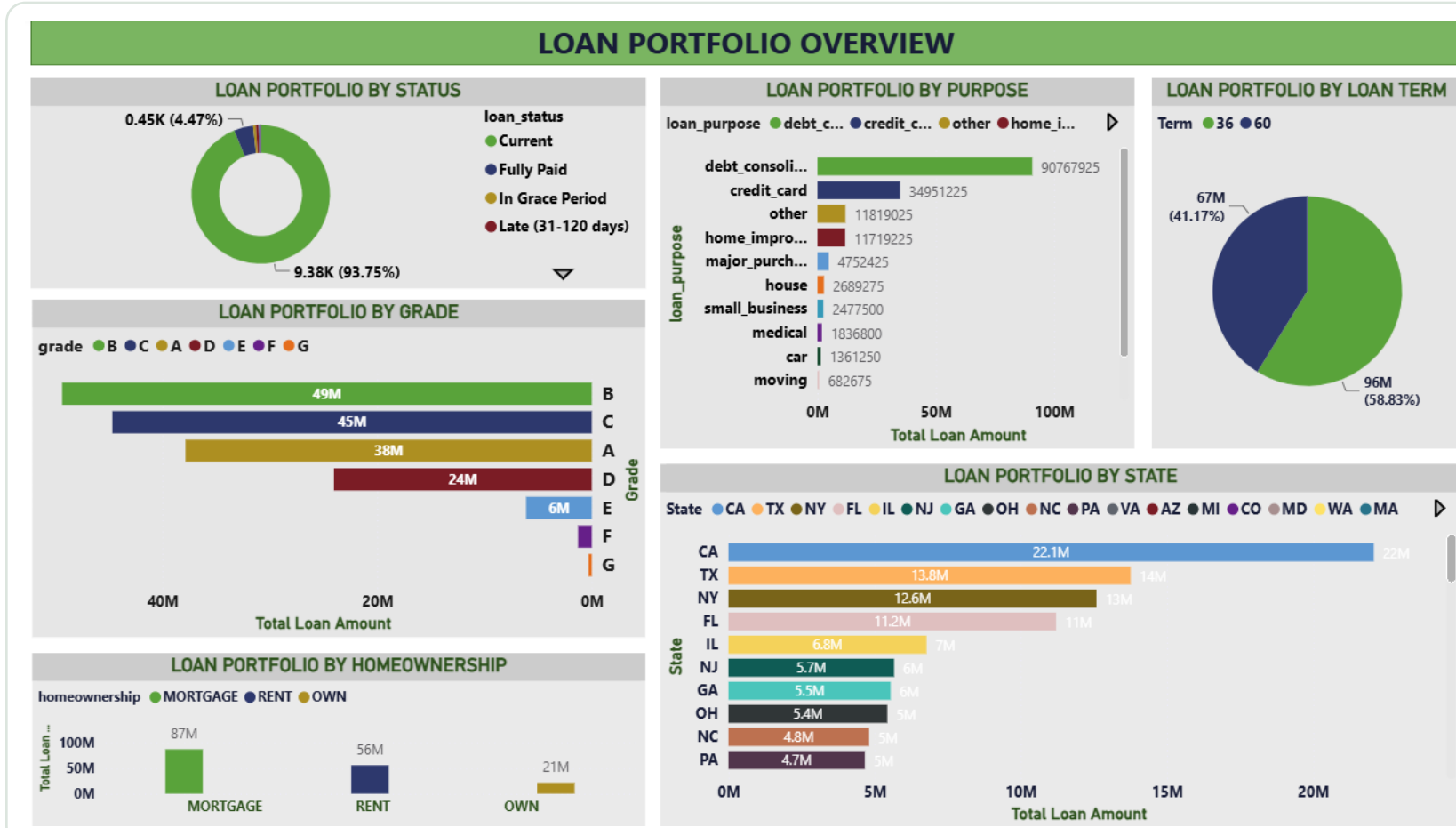
Default Rate

\$164M

Total Exposure

93.75% of loans are current with a 0.07% default rate — debt consolidation and credit cards dominate purpose, and Balanced Logistic Regression best flags high-risk borrowers.

Loan Portfolio Overview

**\$90.7M**

Debt Consolidation

58.83%

60-Month Term Share

\$49M

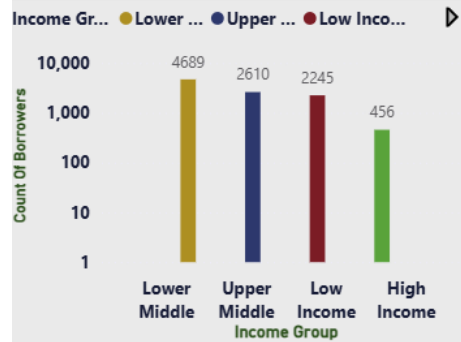
Grade B Exposure

Debt consolidation (\$90.7M) and 60-month terms (58.83%) dominate the book; Grade B is the single largest exposure tier, led by CA, TX, and NY.

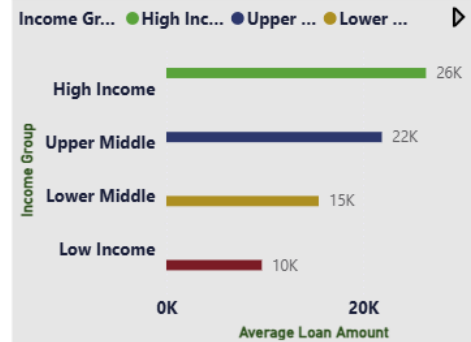
Borrower & Income Analysis

BORROWER AND INCOME ANALYSIS

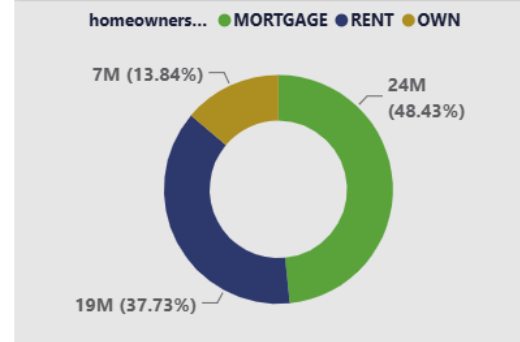
BORROWERS BY INCOME GROUP



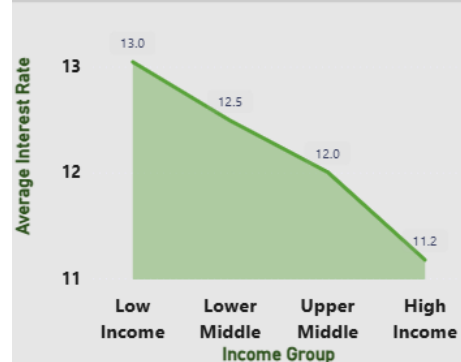
AVERAGE LOAN AMOUNT BY INCOME GROUP



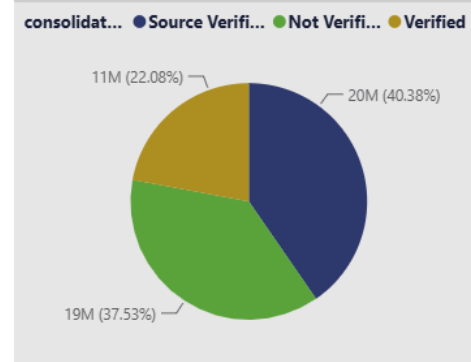
HOMEOWNERSHIP DISTRIBUTION



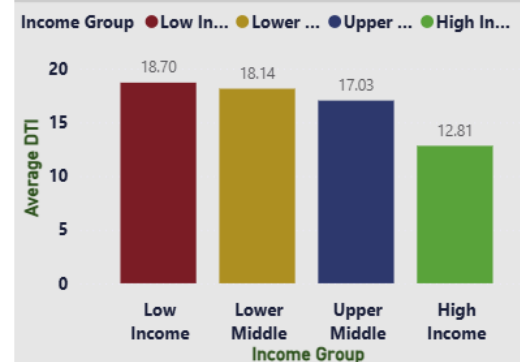
INTEREST RATE BY INCOME GROUP



VERIFIED INCOME STATUS



AVERAGE DTI BY INCOME GROUP

**\$26K**

High-Income Avg. Loan

11.2%

High-Income Avg. Rate

18.70

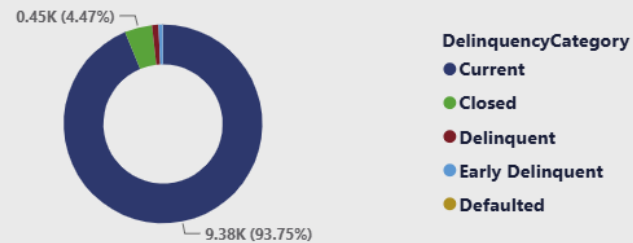
Low-Income Avg. DTI

High-income borrowers get the largest loans (\$26K) at the lowest rates (11.2%), while average DTI falls steadily from 18.70 (low income) to 12.81 (high income).

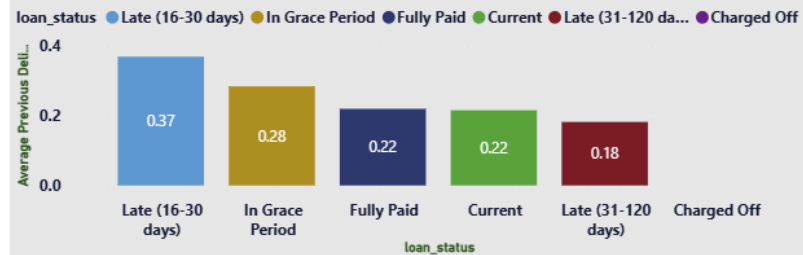
Delinquency & Credit Risk Analysis

DELINQUENCY AND CREDIT RISK ANALYSIS

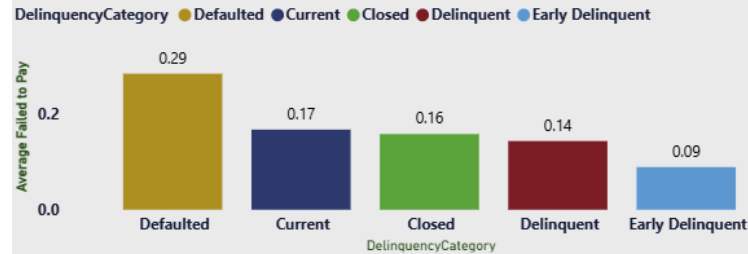
DELINQUENCY CATEGORY DISTRIBUTION



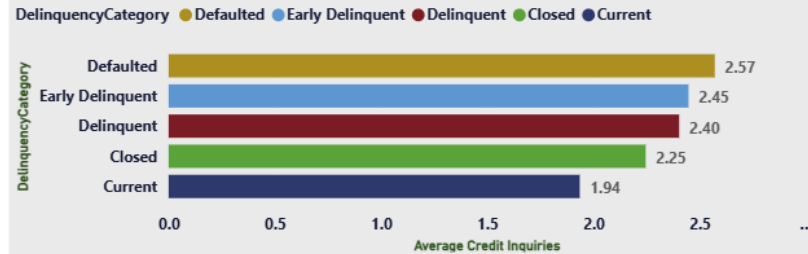
PREVIOUS DELINQUENCIES



HISTORICAL FAILED TO PAY



AVERAGE CREDIT INQUIRY BY DELINQUENCY CATEGORY



AVERAGE TAX LIENS AND AVERAGE BANKRUPTCIES BY DELINQUENCY CATEGORY



0.29

Defaulted Failed-to-Pay

2.57

Defaulted Credit Inquiries

0.14

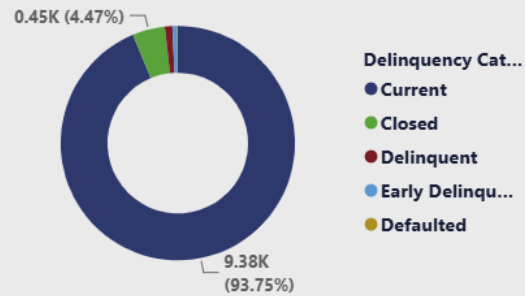
Defaulted Tax Liens

Borrowers who default show the highest failed-to-pay history, the most recent credit inquiries, and the most tax liens & bankruptcies of any delinquency category.

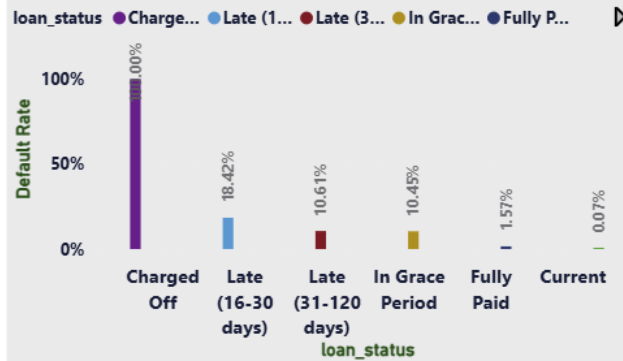
Portfolio Risk Analysis

PORTFOLIO RISK ANALYSIS

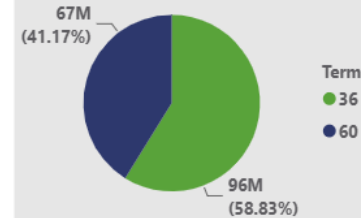
PORTFOLIO HEALTH DISTRIBUTION



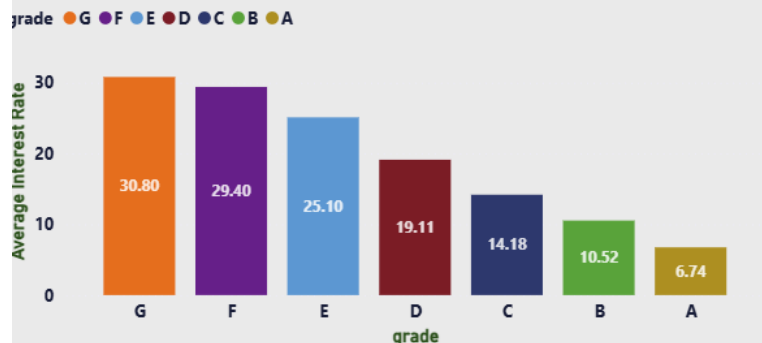
DEFAULT RATE BY LOAN STATUS



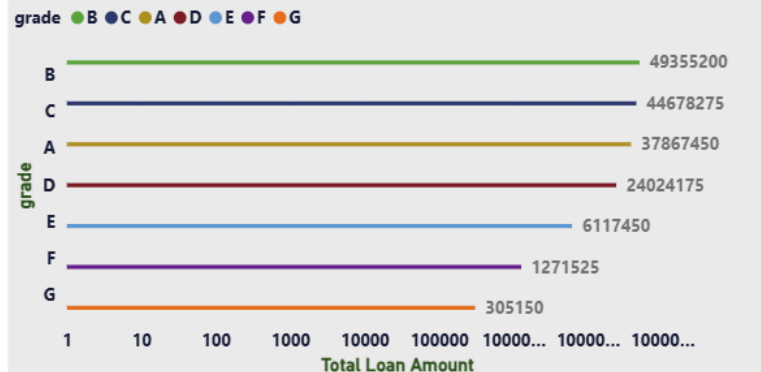
LOAN AMOUNT BY TERM



AVERAGE INTEREST RATE BY GRADE



LOAN EXPOSURE BY GRADE

**18.42%**

Late 16-30d Default Rate

30.80%

Grade G Avg. Rate

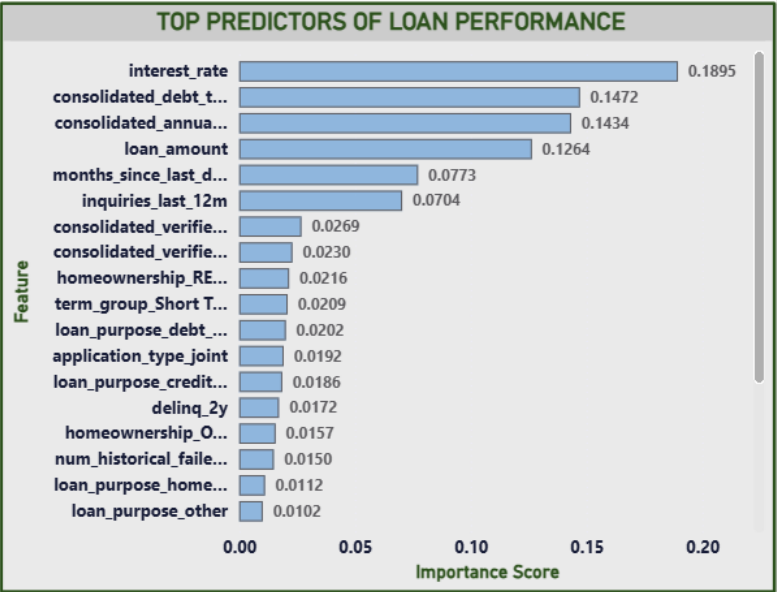
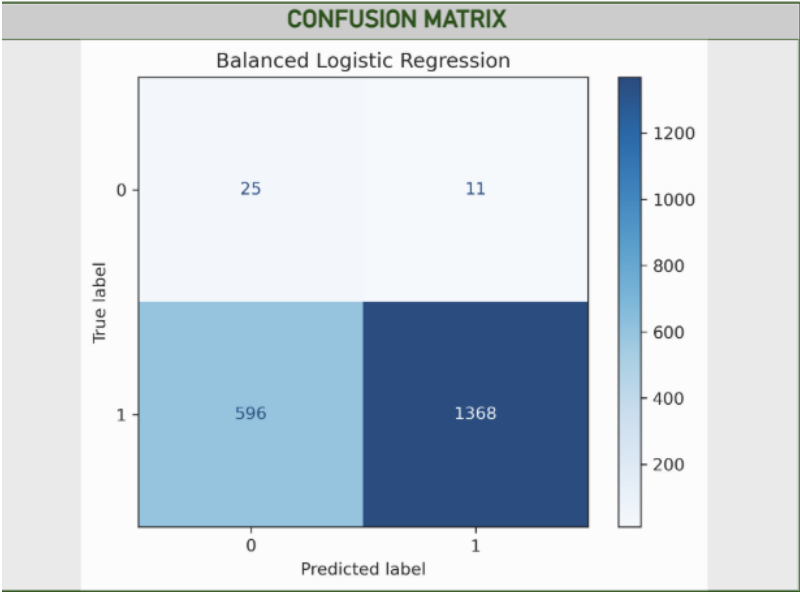
6.74%

Grade A Avg. Rate

Default rate climbs sharply once a loan reaches Late 16-30 days (18.42%); interest rate scales directly with grade, from 6.74% (A) up to 30.80% (G).

MACHINE LEARNING INSIGHTS

MODEL COMPARISON					
MODEL	PRIMARY TASK	ACCURACY	ROC-AUC	NON PERFORMING LOANS DETECTED	RECOMMENDED FOR CREDIT RISK
Balanced Logistic Regression	Classification	0.697	0.762	25 / 36	Yes
Balanced Random Forest	Classification	0.982	0.689	0 / 36	No
Linear Regression	Regression	-	-	-	No
Standard Logistic Regression	Classification	0.982	0.752	0 / 36	No
Standard Random Forest	Classification	0.982	0.681	0 / 36	No



25/36

Non-Performing Caught

0.762

Balanced LogReg ROC-AUC

0.190

Interest Rate Importance

Balanced Logistic Regression trades raw accuracy (69.7%) for real detection — catching 25/36 non-performing loans vs. 0/36 for standard models; interest rate is the top predictor.

BUILT WITH

Excel • SQL • Python • Power BI

This dashboard suite is part of an end-to-end digital lending analytics project covering data cleaning, SQL exploration, statistical analysis, machine learning, and business intelligence reporting.

EA

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